EnergyMonitor Project Charter

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Version Control

Version	Date	Description of change	Authors
1.0	10/16/2026	First version of the project charter: justification, objectives, scope, budget, schedule and team.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.1	10/20/2026	Alignment with the deployed system: account administration in scope, budget with the production server and the estimated total cost, go-live at `https://energy-monitor.tech` with continuous integration and deployment, and status of objectives and milestones as of October 20.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.2	10/24/2026	Scope components according to the architecture in production: users, devices and measurements as independent services behind an API gateway, and migration of the remaining modules in progress.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.3	10/28/2026	Migration completed: the API gateway and the seven services in production, and the monolithic backend retired. Registration with prior email verification and acceptance of the terms, rules for accounts created with Google (no password change and deletion with a code sent by email) and independent editing of the first and last name. Milestone status and final-phase plan as of October 28.	Cortés Parra, J. E.; García Artunduaga, M. A.
1.4	10/28/2026	In-app help (documentation by role, support for users and news that update themselves), profile photos visible in the page and in homes, unusual consumption alert for a device and an explanation of push in Brave.	Cortés Parra, J. E.; García Artunduaga, M. A.

Abstract

This document is the project charter of EnergyMonitor, a solution that lets households see the electrical consumption of each appliance, identify the periods of highest demand, receive alerts and get saving recommendations. It defines why the project is being done and the value it provides, its objectives with measurable indicators, the scope included and excluded, the budget, the phases, the milestones and the schedule until the delivery date, October 28, 2026, and the work team with its roles. The solution integrates a measuring device that attaches to each appliance, a web application and an Android mobile application. As of October 28, 2026 the system is deployed in production at `https://energy-monitor.tech`, split into an API gateway and seven independent services, and is updated through continuous integration and deployment. The detail of the functions is developed in the requirements specification.

Keywords: project charter, electrical consumption, energy saving, households, alerts

Introduction and Justification

Project Context

EnergyMonitor lets households see the electrical consumption of each appliance, identify the periods of highest demand, receive alerts and get recommendations to save energy. Measurement is done by a purpose-built device that attaches to the appliance and sends real readings of its consumption. Users consult the data from a web application and an Android mobile application.

The project is a formative project of the Technologist in Software Analysis and Development (ADSO) program at SENA, developed by two trainees and due on October 28, 2026.

Problem Addressed

High energy consumption in Colombian households is largely caused by a lack of energy awareness. People do not know which appliances consume the most or how their consumption varies throughout the day, so they cannot correct habits such as leaving a device on unnecessarily or keeping an appliance that consumes more than it should. The result is high bills and a greater environmental impact.

Justification

The project starts from the premise that most households have no tools that show their consumption in real time or identify which appliances weigh most on the bill. A system that measures the consumption of each appliance, warns when reasonable limits are exceeded and proposes saving actions based on measured data supports informed decisions and encourages responsible and sustainable use of electricity. Measuring with a real sensor, rather than estimating, is what allows alerts and recommendations to be based on the actual behavior of each household.

Expected Benefits

- For households: knowing which appliances use the energy, reacting in time to excessive consumption and making saving decisions with their own data, which helps control the cost of the bill.
- For the environment: reducing unnecessary electricity consumption and, with it, its environmental impact.
- For the training program: a complete and demonstrable system, from measurement on a real appliance to the notice on the user's phone.

Project Objectives

General Objective

Develop an energy resource control system that monitors household electrical consumption in real time and identifies the appliances that generate the highest expense, in order to promote efficient and responsible use of electricity.

Specific Objectives

Each specific objective has a measurable indicator and an achievable target within the project timeframe. Table 1 presents them together with their coverage as of October 20, 2026.

Table 1

Specific Objectives, Indicators and Coverage

Code	Objective	Indicator and target	Coverage as of October 20, 2026
OBJ-01	Measure the electrical consumption of each appliance in the home	One reading every 60 seconds of voltage, current, active power and accumulated energy for each connected measuring device	Covered. Measuring device operational
OBJ-02	Show consumption clearly in a web and a mobile application	Consumption summary and history per home on the web, and consultation of homes, alerts and recommendations in the Android application, in Spanish, English, French and Portuguese	Web application deployed in production. Mobile application under development, integrated with the API
OBJ-03	Issue automatic alerts for unusual or excessive consumption	Four alert types (device linked, disconnection, high or critical consumption, daily or monthly limit). A disconnection is detected in at most 150 seconds	Covered
OBJ-04	Encourage responsible habits through recommendations based on measured data	Five saving rules evaluated every hour. The same recommendation is not repeated within 7 days	Covered
OBJ-05	Deliver the system working end to end	Live demonstration, from a physical device measurement to an alert or recommendation on a phone, with no manual intervention, on October 28, 2026	System deployed in production. Live demonstration planned for the October 28 delivery; linking a meter in production is pending evidence

General Scope

In Scope

Table 2 lists what the project builds and the system module responsible for each item.

Table 2

Items in Scope

Item	Description	Module or component
Users and profiles	Registration with prior email verification and acceptance of the terms, sign-in, password recovery, Google sign-in, profile (first name, last name and photo) and account deletion	User and Profile Management Module
Administration	Account lookup, status changes, granting and revoking the administrator role, and notices for administrators	User and Profile Management Module
Homes	Homes, members, owner and member roles, favorites, consumption limits, home deletion and ownership transfer	Home Management Module
Devices	Registration, linking, editing and unlinking of measuring devices	Device Management Module
Measurements	Reception of readings, device status, consumption history and statistics	Measurements and Consumption Module
Alerts	Consumption, connection and unusual consumption alerts of the devices	Alerts Module
Recommendations	Saving suggestions based on measured consumption	Savings Recommendations Module
Notifications	Notices by email and in the browser, in-app notices for administrators, preferences and delivery log	Notifications Module
Web application	Full client for management, history and statistics, with help in Settings (documentation, support and news)	Web application
Mobile application	Android client for consultation and alerts	Mobile application
Measuring device	Physical device that measures the consumption of each appliance and sends the readings	Measuring device

Item	Description	Module or component
Deployment	Publication of the system on a server with HTTPS, and updates with automated tests, approval and rollback	Infrastructure

Since October 28, 2026, all the modules run in production as independent services (security, devices, measurements, homes, notifications, alerts and recommendations), behind a single entry point for the API. The core service that used to hold them all was retired. The technical detail is in the technical proposal.

Out of Scope

Table 3 explicitly lists what the project does not build and the reason.

Table 3

Items Out of Scope

Item	Reason
iPhone application (iOS)	The mobile application is developed for Android only.
Manual entry of consumption from the utility bill	The measuring device records consumption automatically, so manual capture is not provided.
Support and content management from the administration	The current administration covers accounts, statuses, roles and notices; the rest is left for a later version.
Cost estimation in local currency	There is no agreement with an energy provider that supplies tariffs. The system reports energy in kWh.
Remote update of the measuring device's software	The device is updated by connecting it with a cable.
Service for several organizations at once	Each installation serves a single organization.

Assumptions and Constraints

Table 4 gathers the assumptions and constraints that condition the project.

Table 4

Project Assumptions and Constraints

Type	Description
Time	Delivery on October 28, 2026, the date set for the evaluation of group 3145556 in the ADSO program, per the ADSO program calendar for group 3145556.
Budget	Approximately COP 180,000 invested in the components of the measuring device and COP 48,000 in one month of server. The domain was obtained at no cost.
Technology	The tools defined for the formative project are used, detailed in the technical proposal.
Team	Two trainees of the ADSO program.
Regulation	Protection of user information, with reference to Law 1581 of 2012 (Congreso de la República de Colombia, 2012).
Connectivity	Continuous Wi-Fi is assumed where measuring devices are installed.

Budget

Table 5 summarizes the project budget.

Table 5

Project Budget

Item	Amount
Measuring device components	Approximately COP 180,000 (COP 173,997 according to the technical proposal breakdown)
Hosting	COP 48,000: Hostinger KVM 2 VPS contracted for one month (COP 27,900 per month on a yearly contract)
Domain and certificates	No cost: free `energy-monitor.tech` domain and Let's Encrypt certificates
Estimated total project cost	COP 57,021,997, including the development team's equipment and the estimated human resources (details in the technical proposal)

Schedule and Milestones

Main Phases

The project started on May 1, 2026 and was planned in four phases with one-week progress steps. Table 6 describes each phase with its period, goal and exit criterion.

Table 6

Project Phases

Phase	Period	Goal	Exit criterion
1. Foundation and core loop	May 1 to June 30	A user registers, creates or joins a home, registers a device and sees its measurements arrive.	A registered user logs in, creates a home, links a device and sees at least one real measurement recorded.
2. Consumption visibility and alerts	July 1 to August 31	Consumption is visible and the system warns when something is abnormal.	A home exceeding its limit produces a visible alert on web and mobile and a notification is delivered.
3. Recommendations and internationalization	September 1 to 30	Close the product loop with proactive guidance and finish the cross-cutting interface work.	A home with one week of history receives at least one recommendation based on its own consumption, and the interface is verified in more than one language.
4. Consolidation and delivery	October 1 to 28	Leave the system ready for demonstration and defense.	The system is demonstrated live, from a physical device measurement to an alert or recommendation on a phone.

Status as of October 20, 2026: the functions of phases 1 to 3 are built and, during phase 4, the system went live at `https://energy-monitor.tech`, with continuous integration and deployment. On October 24, 2026 the separation of the users, devices and measurements modules into independent services went into production, and on October 28, 2026 the separation was completed with the homes, notifications, alerts and recommendations modules and the retirement of the core service, with no changes for users.

Milestones

Table 7 gathers the project milestones with their estimated date.

Table 7

Project Milestones

Code	Milestone	Estimated date	Status as of October 20, 2026
H-01	Project start	May 1, 2026	Done
H-02	End of phase 1: a registered user links a device and sees a real measurement	June 30, 2026	Done
H-03	End of phase 2: an alert for exceeding a limit, visible on web and mobile with its notification	August 31, 2026	Done
H-04	End of phase 3: recommendation based on the home's own consumption and interface in more than one language	September 30, 2026	Done

Code	Milestone	Estimated date	Status as of October 20, 2026
H-05	Complete bilingual documentation (project charter and requirements specification)	October 12, 2026	In progress
H-06	End-to-end tests and user, technical and installation manuals	October 14, 2026	Completed, except the end-to-end tests in production, which are pending evidence
H-07	Technical and cost report, and rehearsal of the demonstration	October 15, 2026	Completed
H-08	Final delivery and live demonstration	October 28, 2026	Pending
H-09	Go-live at `https://energy-monitor.tech` with continuous integration and deployment	October 19, 2026	Done

Final Phase Plan

Table 8 details the activities of the period between October 10 and 28, 2026, which corresponds to phase 4, with the go-live and the delivery.

Table 8

Phase 4 Activity Plan

Dates	Activities	Deliverable
October 10 to 12	Confirm and record the pending changes. Complete the project charter and the requirements specification (ES and EN)	Bilingual documents in APA 7 format
October 13 and 14	End-to-end tests. User manuals (web and mobile), technical and installation manual	Test plan and evidence. Manuals
October 15	Technical and cost report. Rehearsal of the live demonstration	Technical and cost report
October 16 to 20	Deployment on the production server, GitHub Actions pipelines and implementation and deployment manual	System in production and implementation and deployment manual
Until October 24	Separation of the users, devices and measurements modules into independent services, with a single entry point for the API	Services in production and updated documentation
October 25 to 28	Separation of homes, notifications, alerts and recommendations and retirement of the core service. Registration with email verification and terms, rules for Google accounts, end-to-end tests and production deployment of the gateway and the seven services	The API gateway and the seven services in production, and updated final documentation
October 28	Final delivery and live demonstration	Working system and complete documentation

The security improvement to the devices' credentials has already been applied: the system stores only the hash of each device's key. The reliability improvement in the sending of readings has no date within this plan and is carried out only if time allows. The only pending item of home and device management, which is device history management by linking period, also has no assigned date. The separation into services of the Home, Alerts, Recommendations and Notifications modules, carried out by Miguel Ángel García Artunduaga, was completed and deployed on October 28, 2026.

Work Team and Roles

Table 9 presents who does what in the project. The team is made up of two trainees of the ADSO program.

Table 9

Work Team and Roles

Member	Role	Responsibilities
Juan Esteban Cortés Parra	Project lead (Project Manager), Scrum Master, system data manager and developer	Project planning and follow-up, facilitation of the methodology, design and administration of the system data, information analysis, design and programming
Miguel Ángel García Artunduaga	Analyst, QA and developer	Quality assurance and testing, information analysis, design and programming

References

Congreso de la República de Colombia. (2012, October 17). *Ley 1581 de 2012: Por la cual se dictan disposiciones generales para la protección de datos personales* [Law 1581 of 2012: By which general provisions for the protection of personal data are issued]. Diario Oficial No. 48,587.